CHAPTER 8: FORCE & PRESSURE (MEME EDITION)

NCERT Class 8 Science • Hinglish Storyboard + Meme Templates + Exam Topper Notes

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8.1 FORCE — A PUSH OR A PULL (DHAKKA MAAR YA KHEENCH!)

🎬 **MEME SCENE #1: HERA PHERI — BABURAO, RAJU & SHYAM PUSHING KHATARA TAXI**

"Utha le re baba... mere ko nahi, is bhaari bori ko utha le!" — Baburao Ganpatrao Apte

Desi Hinglish Logic: Socho ground par football chup-chaap padi hai. Jab tak aap usko laat (kick) nahi maroge, woh hilegi kya? Nahi! Cricket mein fielder ball rokta hai, hockey player stick se ball ki direction palat deta hai, aur aap subah-subah fridge ka darwaza kholte ho. Science bolti hai ki duniya ke saare काम—picking, kicking, hitting, lifting, opening, shutting—sirf **2 cheezon** mein aate hain: **PUSH (Dhakka dena)** ya **PULL (Apni taraf kheenchna)**! Isi Push ya Pull ko Physics mein **FORCE** kehte hain.

NCERT Activity 8.1 Situation	Asli Action Kya Hua?	Push / Pull?	Desi Meme Vibe
1. Table par rakhi book hilana	Pushing / Pulling / Lifting	**Both (Push & Pull)**	Padhne ka naatak karna 📚
2. Darwaza kholna ya band karna / Drawer kholna	Pushing (band) / Pulling (kholna)	**Both (Push & Pull)**	*"Daya, darwaza tod do!"* 🚪
3. Kuye (well) se paani ki balti kheenchna	Pulling / Lifting rope upwards	**Pull Only**	*Welcome* movie ka kuya scene 🪣
4. Penalty kick / Batsman hitting ball	Kicking / Hitting / Flicking	**Push Only**	*"Mauka Mauka"* sixer shot 🏏
5. Magnet (Chumbak) ka Khel	Attraction = **Pull** | Repulsion = **Push**	**Both (Push & Pull)**	Dost jab udhaar maange = Repulsion 🧲

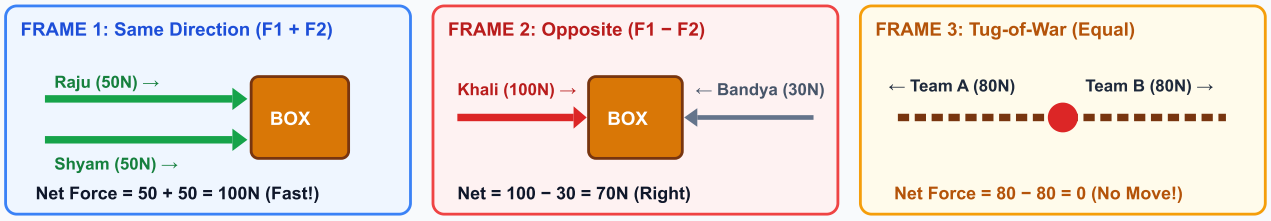
8.2 FORCES ARE DUE TO AN INTERACTION (AKELE FORCE NAHI LAGTA!)

🎬 **Meme Scene #2 (Welcome Movie — Majnu Bhai & Band Padi Red Car):** Man lo aapki gaadi beech sadak par band ho gayi (NCERT Fig. 8.2). Agar aap gaadi ke peeche haath baandh kar khade ho jaoge aur usko ghooroge —*"Chal na bhai!"*—toh kya gaadi chalegi? Bilkul nahi! Gaadi tabhi hilegi jab aapke haath gaadi ki dikki ko **touch karke dhakka (interact)** lagayenge. Chahe do ladkiyan haath pakad kar gol-gol ghoom rahi hon (pulling each other) ya ek aadmi ziddi bail (cow) ko rassi se kheench raha ho (Fig. 8.3)—dono ek-dusre par force laga rahe hain!

🏆 **EXAM TOPPER BOX (CHHAAP DO EXAM MEIN — 100% MARKS)**

- **Definition of Force:** In science, a **push or a pull** acting on an object is called a **force**. Motion is imparted to objects due to the action of a force.
- **Attraction & Repulsion:** Magnetic attraction is a form of *Pull*, while repulsion between similar poles is a form of *Push*.
- **Rule of Interaction:** At least **two objects must interact** with each other for a force to come into play. Without interaction between two objects, no force can act.

8.3 EXPLORING FORCES (EKTA MEIN BAL VS TUG-OF-WAR!)



🎥 3-Frame Animation Strip: Activity 8.2 (Heavy Box) & Fig. 8.5 (Tug-of-War Rassi Kheenchna)

🎬 **Meme Scene #3 (Chup Chup Ke — Bandya vs Heavy Almari):** Jab aap akele bhaari almari dhakelte ho toh kamar toot jaati hai. Par jab aapka dost bhi **same direction** mein dhakka lagata hai (Fig. 8.4a), toh dono ki taakat jud jaati hai ($F_{net} = F_1 + F_2$)! Lekin agar aapka harami dost **opposite side** se dhakka marne lage (Fig. 8.4b), toh fight ho jaati hai aur badi force mein se choti force minus hoti hai ($F_{net} = F_1 - F_2$). Aur Tug-of-War (rassi kheenchna) mein jab dono teams barabar dum lagati hain, toh rassi beech mein hi atki rehti hai kyunki **Net Force = ZERO!**

8.4 A FORCE CAN CHANGE THE STATE OF MOTION (SPEED & DIRECTION)

🎬 **MEME SCENE #4: PANCHAYAT — "DEKH RAHA HAI BINOD, TYRE KAISE BHAGAYA JAA RAHA HAI!"**

"Chalti gaadi ko dhakka doge toh Rocket banegi, aur aage se rokoge toh Brake lagega!"

Desi Hinglish Logic (Activities 8.3 & 8.4): Bachpan mein dande se cycle ka purana rubber tyre bhagaya hai (Fig. 8.7)? Jab tyre chal raha hota hai aur aap usi direction mein danda maarte ho, toh uski **speed badh jaati hai**. Lekin jab football goalkeeper chalti ball ke samne haath lagata hai (opposite direction), toh ball ki **speed kam ya zero** ho jaati hai! Aur agar chalti ball ke raaste mein tedha scale (ruler) rakh do ya Virat Kohli bat se flick maare (Fig. 8.8), toh ball ki **direction hi badal jaati hai!**

⚠️ *Meme Twist:* Kya force lagane se hamesha motion badalta hai? Apne kamre ki deewar (wall) ko poori taakat se dhakka maaro—deewar hili kya? **Nahi!** (Kyunki force kaafi nahi hai).

🏆 **EXAM TOPPER BOX (8.3 & 8.4 FORMULAS & DEFINITIONS)**

- **Magnitude & Direction:** The strength of a force is expressed by its **magnitude**. Force has both magnitude and direction.
- **Net Force Rules:** (1) Same direction: forces add up ($F_1 + F_2$). (2) Opposite direction: net force is the difference ($F_1 - F_2$), acting in the direction of the larger force. (3) Equal & opposite forces: **Net Force = 0**.
- **State of Motion:** Described by an object's **speed** and **direction of motion**. Rest is the state of **zero speed**. A change in speed, direction, or both is called a **change in the state of motion**.

8.5 FORCE CAN CHANGE THE SHAPE OF AN OBJECT (MUMMY KA BELAN!)

🎬 **MEME SCENE #5: TMKOC JETHALAL — AATE KI LOI (DOUGH) VS BELAN ATTACK!**

"Goli beta, masti nahi! Toothpaste ko dabaya toh bahar aaya, aur gubbare ko dabaya toh pichak gaya!"

Desi Hinglish Logic (Activity 8.5): Jab thaali mein gundha hua aata (dough) rakha hota hai aur aap usko mukka maarte ho ya belan chalaate ho, toh aata bhaag kar kamre se bahar nahi jaata—uska **SHAPE badal kar gol chapati** ban jaata hai! Waise hi cycle ki gaddi par baithne se spring dab jaati hai, aur do eenton (bricks) ke beech rakhe plastic scale par wazan rakhne se scale beech se mud (bend) jaata hai. Yaani **Force ke कुल 5 Maha-Kaam hain**: (1) Rest se move karana, (2) Speed badhana/ghatana, (3) Direction badalna, (4) Shape badalna, ya (5) Inmein se sab kuch ek saath!

NCERT Table 8.2 Object	Force Kaise Lagaya?	Change in Motion?	Change in Shape?
1. Lump of dough (Gundha aata) on plate	Haath se neeche dabana (Pressing down)	**No** (Wahi rehta hai)	**YES** ✅ (Chपटा ho gaya)
2. Spring fixed to bicycle seat	Seat par baithna (Sitting on seat)	**No**	**YES** ✅ (Compress hua)
3. Rubber band hanging on wall hook	Neeche wazan latkana ya kheenchna	**No**	**YES** ✅ (Stretch/Lamba)
4. Plastic scale between two bricks	Beech mein weight rakhna	**No**	**YES** ✅ (Bend ho gaya)

8.6 CONTACT FORCES: MUSCULAR FORCE & FRICTION (CHHUO TOH JAANE!)

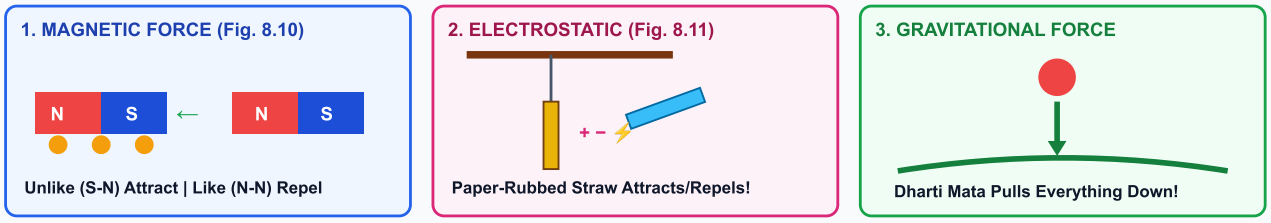
1. Muscular Force (Dole-Shole Ki Taakat): Kya aap table par rakhi paani ki balti ko aankhon se ghoor kar bina chhue utha sakte ho? Nahi, hum *Stranger Things* ki Eleven nahi hain! School bag uthane, football kick karne, yahan tak ki **saans lene (lungs expand/contract)** aur **khana pachane (alimentary canal mein food push karne)** mein bhi hamari body ke muscles kaam karte hain. Bail (bullocks), ghode aur oont (camels) bhi gaadi kheenchne mein Muscular Force lagate hain (Fig. 8.9). Kyunki yeh bina touch kiye nahi lagti, isliye ise **Contact Force** kehte hain.
2. Force of Friction (Mirzapur Kaleen Bhaiya — "Roko Isko!"): Socho cycle ke pedal maarna band kar do, toh cycle apne aap kyun ruk jaati hai? Ground par ludhakti ball apne aap kyun thak kar ruk jaati hai? Kyunki zameen aur tyre/ball ke beech ek chhupa rustam **Friction Force** lagta hai jo **hamesha motion ke ULTE (opposite) direction** mein taang kheenchta hai!

🏆 **EXAM TOPPER BOX (8.5 & 8.6 GOLDEN POINTS)**

- **Contact Force:** A force that acts on an object **only when it is in physical contact** with it (directly or via a stick/rope).
- **Muscular Force:** The force resulting due to the **action of muscles** in humans and animals (e.g., lifting, chewing, digestion, breathing).
- **Friction:** A contact force between two surfaces in contact that **always opposes motion** (acts opposite to the direction of motion).

8.7 NON-CONTACT FORCES (BINA CHHUE JADOO — WIRELESS FORCES!)

🎬 **Meme Scene #6 (Welcome Movie Uday Shetty — "Control Uday, Control!"):** Abhi tak humne dekha ki dhakka dene ke liye chhuna padta hai. Lekin Physics mein **3 aisi Superpowers (Non-Contact Forces)** hain jo Wi-Fi aur Bluetooth ki tarah **bina touch kiye** door se hi cheezon ko kheench ya dhakel deti hain!



🎥 3-Panel Storyboard: Magnetic Rollers (Act 8.6), Charged Straw (Act 8.7) & Gravity Pull

- 🧲 **1. Magnetic Force (Activity 8.6 — Pencils Par Magnet Ka Dance):** Jab aap 3 gol pencils ke upar ek bar magnet rakhte ho aur dusra magnet paas laate ho (bina touch kiye!), toh pencil wala magnet apne aap aage-peeche bhaagne lagta hai! *Rule:* **Like poles (N–N ya S–S) Repel** (door bhaag), aur **Unlike poles (N–S) Attract** (paas aao). Lohe (iron) ko kheenchna bhi non-contact magnetic force hai.
- ⚡ **2. Electrostatic Force (Activity 8.7 — Plastic Straw Ka Jadoo):** Bachpan mein sukhe baalon mein plastic scale ragad kar kaagaz ke tukde uthaye hain? NCERT Activity 8.7 mein ek plastic straw ko dhaage se latkao. Dusre straw ko kaagaz (paper) se ragad kar paas lao—woh latke hue uncharged straw ko **attract (kheench)** lega! Lekin agar **dono straws ko paper se ragad** doge, toh dono par same charge aa jayega aur woh ek-dusre se **door bhaagenge (Repel)**!
- 🌍 **3. Gravitational Force (Newton Ka Seb & Behti Nadi):** Haath se coin chhoote ya ped se aam toote—sab neeche hi kyun girta hai? Kyunki **Earth har cheez ko apni taraf pull karti hai**, jise **Force of Gravity** kehte hain! Nal kholte hi paani ka neeche girna aur nadiyon (rivers) ka behna sab Gravity ki wajah se hai. *Big Fact:* Sirf Earth hi nahi, ब्रह्मांड (Universe) ka har chota-bada object har dusre object ko attract karta hai—ise **Gravitational Force** kehte hain!

🏆 **EXAM TOPPER BOX (8.7 NON-CONTACT DEFINITIONS)**

- **Non-Contact Force:** A force that comes into play even when the objects are **not in physical contact** with each other.
- **Electrostatic Force:** The force exerted by a **charged body** on another charged or uncharged body without contact.
- **Gravity vs Gravitation:** Earth's attractive pull on objects is called **gravity**. The universal attractive force between any two objects in the universe is called **gravitational force**.

8.8 PRESSURE = FORCE ÷ AREA (CHOTA AREA, BADA DHAMAKA!)

🎬 **MEME SCENE #7: SHARK TANK ASHNEER GROVER — "BHAI KYA KAR RAHA HAI TU? NUKILI KEEL THOK NA!"**

Formula: Pressure = Force ÷ Area (Yaani Area Jitna CHOTA, Pressure Utna ZYADA!)

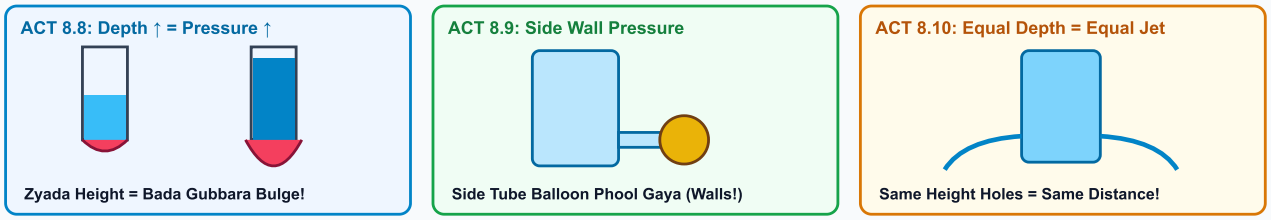
Desi Hinglish Logic (3 Famous Exam Questions):

1. Keel (Nail) & Chaku (Knife) (Fig. 8.12): Deewar mein keel (nail) thokte waqt uska nukila (pointed) hissa andar kyun rakhte hain? Kyunki pointed tip ka **Area bahut chota** hota hai, toh thode se Force se bhi **bahut bada Pressure** lagta hai aur keel lakdi चीर kar andar chali jaati hai! Same logic tez dhaar wale chaku (sharp knife) mein lagta hai.

2. Coolie (Porter) Ki Gol Pagdi (Fig. 8.13): Railway station par Coolie bhaari peti uthane se pehle sar par gol kapda (pagdi) kyun rakhta hai? Taaki sar ke saath bhoj ka **Area of contact BADH jaye**, jisse sar par **Pressure KAM ho jaye** aur dard na ho!

3. School Bag Ki Chodi Pattiyan (Broad Straps): Bag ki straps chodi isliye hoti hain taaki Area zyada ho aur kandhon par Pressure kam pade!

8.9 PRESSURE EXERTED BY LIQUIDS AND GASES (PAANI AUR HAWA KA ZOR!)



🎥 **Liquid Pressure Storyboard: Fig. 8.14 (Bottom), Fig. 8.15 (Side Walls) & Fig. 8.16 (Equal Depth)**

Desi Hinglish Logic (Liquids & Gases):

• **Activity 8.8 (Fig. 8.14):** Pipe ke neeche gubbara baandh kar paani dalo—jitna **zyada paani (height/depth)** bharoge, neeche ka gubbara utna zyada phoolega!

• **Activity 8.9 & 8.10 (Fig. 8.15 & 8.16):** Bottle ki side mein tube lagane par bhi gubbara phoolta hai, aur bottle ke charon taraf **same height par chhed (holes)** karne par paani ki dhaar **barabar doori (equal distance)** par girti hai! Matlab paani walls par bhi pressure lagata hai aur same depth par barabar pressure lagata hai.

• **Gases Bhi Kam Nahi:** Gubbare mein hawa bharne par woh charon taraf se phoolta hai, aur chhed ho ya cycle tube puncture ho toh hawa bhaag jaati hai—kyunki **gases bhi container ki walls par har direction mein pressure lagati hain!**

🏆 **EXAM TOPPER BOX (8.8 & 8.9 GOLDEN RULES)**

- **Definition of Pressure:** The force acting on a **unit area** of a surface is called pressure (**Pressure = Force ÷ Area**). Smaller area = larger pressure for the same force.
- **Three Laws of Liquid & Gas Pressure:** (1) Pressure at the bottom depends on the **height of the liquid column**. (2) Liquids and gases exert pressure on the **walls of their container**. (3) Liquids exert **equal pressure at the same depth**.

8.10 ATMOSPHERIC PRESSURE (SAR PAR 225 KG KA BOJH AUR HUM ZINDA HAIN?!)

🎬 **MEME SCENE #8: 3 IDIOTS VIRUS SHOCKED — RUBBER SUCKER, 225 KG HEAD LOAD & 16 GHODE!**

"Sar par 225 Kilo (2250 N) hawa ka bojh hai, phir bhi hum pichakte kyun nahi? Kyunki andar ka khoon barabar ulta dhakka maarta hai!"

1. Kya Hai Atmospheric Pressure (Fig. 8.17): Hamare charon taraf kai kilometres upar tak hawa ki chadar (**Atmosphere**) hai. Is hawa ke wazan se jo pressure lagta hai use **Atmospheric Pressure** kehte hain.
2. Rubber Sucker Experiment (Activity 8.11, Fig. 8.18): Jab rubber sucker ko table par dabate hain, toh andar ki hawa nikal jaati hai aur bahar ka BADA Atmospheric Pressure usko daba kar chipka deta hai!
3. 15 cm × 15 cm Sar & 16 Horses (Fig. 8.19 & 8.20): Hamare **15 cm × 15 cm** ke sar par hawa **225 kg (2250 N)** ke barabar force lagati hai! Hum isliye nahi pichakte kyunki **hamari body ka internal pressure bahar ke atmospheric pressure ko balance** kar deta hai. 17th century mein German scientist **Otto von Guericke** ne 51 cm ke do khokhle metal katoron (hemispheres) se pump dwara hawa nikaal di—toh dono taraf **8-8 ghode (total 16 horses)** milkar bhi unhe alag nahi kar paaye!

📝 COMPLETE NCERT EXERCISES SOLVED (Q1 TO Q10 RAPID FIRE!)

NCERT Q. No.	100% Exam-Ready Answer (With Desi Memory Hook)
Q1 & Q2 (Examples)	**Q1 (Change Motion):** *Push:* Kicking a football, pushing a heavy box. *Pull:* Drawing a bucket from a well, pulling a drawer. **Q2 (Change Shape):** Rolling chapati dough with a belan; pressing an inflated balloon between palms.
Q3 (a, b, c, d) (Fill Blanks)	(a) To draw water from a well we have to **pull** at the rope. | (b) A charged body **attracts** an uncharged body. (c) To move a loaded trolley we have to **push or pull** it. | (d) North pole of a magnet **repels** the north pole of another.
Q4 (a, b, c, d) (Archer Bow)	(a) Change in its **shape**. | (b) Example of **muscular** force. | (c) Example of a **contact** force. (d) Forces acting on moving arrow are due to **gravity** and due to **friction** of air.
Q5 (a, b, c, d) (Agent & Effect)	**(a) Lemon:** Agent = Fingers (muscular), Object = Lemon, Effect = Change in shape (juice out). **(b) Toothpaste:** Agent = Fingers, Object = Tube, Effect = Shape change. **(c) Spring load:** Agent = Suspended load, Object = Spring, Effect = Elongation (shape change). **(d) High jump athlete:** Agent = Athlete's muscles pushing ground, Object = Athlete's body, Effect = Change in state of motion.
Q6 & Q7 (Blacksmith/Balloon)	**Q6 (Blacksmith hammering hot iron):** Muscular force of hammering **changes the shape** of the hot iron piece into a tool. **Q7 (Rubbed balloon sticks to wall):** Due to **Electrostatic Force** acquired by rubbing the balloon with synthetic cloth.
Q8, Q9 & Q10 (Bucket, Rocket & Dropper)	**Q8 (Bucket held in hand):** Downward *Gravity* + Upward *Muscular Force*. They are equal and opposite, so **Net Force = 0** (no motion). **Q9 (Rocket fired upwards):** (1) Downward **Force of Gravity** and (2) Downward **Friction of air**. **Q10 (MCQ — Water rising in dropper):** Correct Option = **(d) atmospheric pressure** (pushes water inside low-pressure bulb).

- 💡 **Bonus Extended Learning Trick (Fig. 8.21 Ulta Glass & Card):** Pani se bhare glass par postcard rakh kar ulta karne par card nahi girta kyunki **neeche se lagne wala upward Atmospheric Pressure** glass ke andar ke paani ke downward pressure se zyada hota hai!